# Mateti Harshan

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## EDUCATION

- **Keshav Memorial Institution of Technology** — Hyderabad, India
  *Bachelor of Technology ; GPA: 8.66* — *Aug 2023 - June 2027*
  ***Courses:*** *Operating Systems, Data Structures, Analysis Of Algorithms, Artificial Intelligence, Machine Learning, Networking, Databases*

## SKILLS SUMMARY

- **Languages**: C++, Python, JavaScript, SQL, JAVA, C, MongoDB
- **Frameworks**: Scikit, NextJS, TensorFlow, Keras, Flask, NodeJS
- **Tools**: Kubernetes, Docker, GIT, MongoDB, MySQL, SQLite , Auto Docking Vina , Rdkit
- **Platforms**: Linux, Web, Windows, Arduino, AWS Cloud
- **Soft Skills**: Leadership, Event Management, Writing, Public Speaking, Time Management
- **AI / Robotics**: Machine learning, Generative AI, Neural Networks , Large language models , Edge Impulse, Open CV

## PROJECTS

- **GenAI – Insulin Drug Analysis**: Analyzed insulin drug datasets for diabetes treatment using pharmacological constraints such as molecular weight, solubility, bioavailability, hydrogen bonding, and drug-likeness metrics to generate refined datasets and analytical reports. Reference : Review 1 Review 2
- **Adobe India Hackathon – Connecting the Dots Challenge**: Expanded offline, CPU-optimized PDF intelligence systems including a hierarchical outline extractor (Title–H3 with page numbers) and a persona-driven document analysis engine to rank contextually relevant sections across multiple PDFs. Accomplished high-performance processing (<10 seconds for 50-page PDFs) with a scalable, privacy-preserving planned .Reference : Review
- **TabulaX – Leveraging Large Language Models for Multi-Class Tabular Transformation**: Designed an LLM-powered framework to transform heterogeneous tabular data by classifying tables into string-based, numerical, algorithmic, and general types. Generated interpretable transformation functions and realized higher accuracy than state-of-the-art methods on diverse datasets.Reference : Document review 1 review 2
- **EduX – AI-Enabled Educational Platform**: Built a full-stack learning platform with dynamic quiz generation, student performance analytics, and adaptive feedback. Integrated YOLOv5 for real-time phone detection during assessments to ensure academic integrity. Reference : Review
- **AI-Powered Multimodal Insurance Fraud Detection System (MEGATHON 2025 – IIIT Hyderabad)**: Built an end-to-end AI-driven car insurance claim validation system for CHUBB Insurance within 15 hours, integrating Computer Vision, NLP, and predictive analytics. Implemented damage detection, part localization, severity estimation, cost prediction, and multimodal fraud detection using CNNs, BERT, and explainability techniques (Grad-CAM) to enable transparent and automated claim verification. Reference : Review
- **IoT-Based Vehicle Automation System**: Improved a voice-enabled web application to control vehicle functions, integrating live location tracking, weather prediction, and music streaming using external APIs with real-time data storage and authentication.Reference : Review

## HONORS AND AWARDS

- Contributed for Chubb Insurance Company in MEGATHON 2025 @ IIIT Hyderabad
- Contributed as official Contributor for the Plone Volto in Google Summer of Code , 2024
- Runner's Up at Adobe India Hackathon – Connecting the Dots Challenge

## RELEVANT EXPERIENCE

- **Adobe India -Connecting the Dots Challenge** — Hyderabad, India
  *Designed an **NLP-driven multi-PDF analysis pipeline** that extracts persona-specific insights from large document collections, **reducing manual document review time by 70%**.* — *Jun -2025*
- **Drug Design** — Hyderabad, India
  *Enhanced an AI-driven drug design pipeline to generate and evaluate **1,000+ insulin compound candidates** using pharmacological descriptors, achieving **98% prediction accuracy** in compound effectiveness classification.* — *Aug - 2025*